

Regression Test Log

ISSUE-D7-02 — FIFO Multi-Wrap Pointer Test

Purpose

Confirm pointer wraparound arithmetic is correct across *multiple consecutive* wraps, not just the single wrap the original Day 6 test exercised — a systematic addressing error at the wrap boundary could plausibly only surface on a second or third wrap if the first one happened to mask it.

Test

`tb/sync_fifo_multiwrap_tb.sv` runs 3 consecutive fill-to-8-then-drain-to-0 cycles back to back (24 writes, 24 reads total), using a different value range each loop (`0x00-0x07`, `0x20-0x27`, `0x40-0x47`) so a stale-data bug from a previous wrap would be immediately obvious.

Every read is checked automatically against a software reference queue (a scoreboard, built for this regression test specifically) rather than relied on manual inspection.

Bug caught and fixed while building this test, before it ever ran against the DUT. The first draft of this testbench had `do_write` and `do_read` using inconsistent wait patterns (wait-then-set vs. set-then-wait), which would have silently dropped the last write of every fill phase — it would never get clocked in before being overwritten by the next task. Caught by reasoning through the task sequencing before trusting any result, not by the simulation itself: a task-sequencing bug in a testbench doesn't announce itself the way a compile error does.

Result

Verbatim.

```
Running 3 consecutive fill-then-drain cycles (24 writes, 24 reads total)...  
PASS: all 24 reads across 3 wraps matched the reference queue exactly.  
Multi-wrap regression test complete.
```

Verdict

Zero mismatches across 24 reads spanning 3 full wraps of both `wr_ptr` and `rd_ptr`. Pointer wraparound arithmetic (`((ptr == DEPTH-1) ? 0 : ptr+1)`) is confirmed correct under repeated use, not just a single pass.

ISSUE-D7-02 is resolved.